2

Something Rotten in the Eigenstate of Denmark

The call had finally come for Werner Heisenberg. The fresh-faced physicist, all of twenty-four years old, had been invited to give a talk at the University of Berlin, the center of physics in Germany and arguably the world. He would be explaining his astonishing new ideas in front of Einstein himself.

"Since this was my first chance to meet so many famous men, I took great care to give a clear account of the concepts and mathematical foundations of what was then a most unconventional theory," Heisenberg recalled decades later. "I apparently managed to arouse Einstein's interest, for he invited me to walk home with him so that we might discuss the new ideas at greater length."

As they walked to his apartment on that spring day in 1926, Einstein innocuously asked Heisenberg about his education and background, careful not to turn the subject to Heisenberg's new theory. He waited until they were safely indoors before he sprung the trap.

Heisenberg's "most unconventional theory" was an enormous breakthrough. It promised to solve the outstanding scientific challenge of